Detailed Report

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Data Preparation and Cleaning Documentation

1. Import csv. 43236 obs and 212 variables.
2. Inspected dimensions and found out that the first 3 columns did not have relevant information. However, did not begin any data cleaning process yet.
3. Manually verifying all 212 columns would be time-consuming so referred to dictionary for the essential columns
4. Checked the following 6 columns - loan_amnt", "funded_amnt", "loan_status", "recoveries", "collection_recovery_fee", recovery_rate", "CCF"
5. Every single datapoint in the csv is a charged off loan, which means recovery was no complete. 42475 are charged off, and 761 are Did not meet credit policy : Charged off.
6. Inspected the 'summary' (mean, median etc) for loan amount, funded amount, recoveries, recovery rate and CCF.

Variable	Key result	Interpretation
loan_amnt	Mean = 14,431; median = 12,375	The average requested loan amount is higher than the median, suggesting some larger loans pull the average upward.
funded_amnt	Mean = 14,378; median = 12,300	The amount actually funded is slightly lower on average than the amount requested.
recoveries	Mean = 920.40; median = 150.20	Recoveries are strongly right-skewed: a relatively small number of large recoveries pull up the mean.
recovery_rate	Mean = 6.08%; median = 2.95%	The typical recovery rate is low, with substantial variation across loans.
CCF	Mean = 0.736; median = 0.790	Values range from approximately 0.0004 to 1. The variable's definition still needs to be established.

7. No recoveries is negative
8. There is a data point with recovery rate of 1. Need to check this. How is the csv file calculating the recovery rate? Based off of the loan amount or funded amount?
9. Recovery rate calculated based on 'Loan amount' has 281 mismatches and based on 'funded amount' has 27 mismatches. So there is no consistent formula being used for the recovery rate.
10. After inspecting the 27 mismatches for recovery rate calculated using funded amount, we find that the recovery exceeded the funded amount, so the recover rate > 1; but was reported as 1. Since LGD = 1 – recovery rate, the maximum the recovery rate can be classified as 1, otherwise we will get negative recovery rates.

Check	Result	Interpretation
Mismatches with supplied recovery_rate = 1	27	All 27 have a supplied rate of 100%.
Mismatches with calculated rate above 1	27	All 27 have raw recoveries exceeding the funded amount.
Minimum calculated rate among mismatches	1.010920	Recoveries are approximately 101.09% of funded amount.
Maximum calculated rate among mismatches	1.220774	Recoveries are approximately 122.08% of funded amount.

11. Out of the 43236 observations, 42860 have same loan amount and funded amount. 376 have a difference. For the 27 mismatch records, the amounts DID NOT differ. Which means a different formula is being used for different data points.
12. Credit Conversion Factor, CCF is present in the dataset but not defined in the dictionary. Need to verify if the columns is actually CCF or something else.

Variable compared with CCF	Correlation	Interpretation
loan_amnt	0.1911	Weak positive relationship
funded_amnt	0.1926	Weak positive relationship
recoveries	0.2152	Weak positive relationship
recovery_rate	0.2007	Weak positive relationship

13. CCF does not seem to move along any of the above values. So need to check what is represents. Tested with some formula hypothesis and did not match.
14. Found online that CCF formula is (funded amount – principal received)/funded amount. Verified that it matches. There was 0 mismatch and the first 10 values exactly coincided.
15. Inspect for missing values

Measure	Result	Meaning
Total rows	43,236	Loans in the dataset
Total columns	212	Original variables
Columns with missing values	22	Need further assessment
Completely empty columns	17	No usable observations
Columns with no missing values	190	Fully observed
Total missing cells	791,936	Missing entries across the dataset

16. Inspecting the 5 columns (22 missing – 17 all empty) with partially missing data.

Variable	Missing	Initial assessment
mths_since_last_major_derog	81.61%	High missingness; investigate before deciding whether to retain
tot_coll_amt	24.93%	Potentially useful; investigate missingness and predictive relevance
tot_cur_bal	24.93%	Potentially useful; investigate missingness and predictive relevance
revol_util	0.12%	Very little missingness; imputation may be straightforward
collections_12_mths_ex_med	0.06%	Very little missingness; imputation may be straightforward

17. We need to check what type of missingness are we dealing with. For the missing data, is there any relation with the recoveries? Or is there no effect?

Group	Loans	Positive-recovery rate
tot_cur_bal observed	32,456	50.25%
tot_cur_bal missing	10,780	74.78%

For ‘tot_cur_balance’ we see that the missing data corresponds to datapoints that tend to have a higher recovery rate. Which means the missing data has relevant information.
After doing a z-test we got p value close to 0. This means that the missing data is statistically significant. We cannot just drop those rows or impute them without using relevant techniques.

18. Checking the same for tot_coll_amt

Group	Loans	Positive-recovery rate
tot_coll_amt observed	32,456	50.25%
tot_coll_amt missing	10,780	74.78%

The result is exactly the same. Which means the two columns seem to be highly correlated. They also show the same pattern, which means that the missingness has relevant information to help without LGD predictions.

19. After doing the same check on 'revol_until' and 'collections_12_mths_ex_med', we can say that the missingness is very little. Rather than impute with 0 we can use median perhaps.
20. Checked all empty columns. They have no contribution to our model and zero data. There are 17 such columns.
21. Before proceeding we check for all the different data types present within the dataset.

Data type	Number of columns	Meaning
Character	24	Text, categories, dates, identifiers
Integer	137	Whole-number values
Logical	17	TRUE/FALSE values
Numeric	34	Decimal or continuous numerical values
Total	212	All original columns

The 17 logical columns are most likely the empty columns being shown as Boolean for 0. This was confirmed to be true.

22. There are 0 duplicate rows, and all loan ID are unique.
23. The loan status

Loan status	Number of loans	Percentage
Charged Off	42,475	98.24%
Does not meet the credit policy: Charged Off	761	1.76%
Total	43,236	100.00%

All loans are defaulted loans. There are no loans that are non-default.

24. Recovery outcomes

Recovery outcome	Loans	Percentage
Positive recovery	24,371	56.37%
Zero recovery	18,865	43.63%
Total	43,236	100.00%

Assumption: for the zero recovery, it is being assumed that the data is correct and that there was no recoveries for the specific loan. We are not assuming that the data is missing and 0 is a placeholder.

25. There is a column called good_bad where every single data entry is 0. Note, this is not one of the 17 missing columns.
26. One hot encoders

Variable	Audit conclusion	Modelling decision
loan_status	Only two encoded indicators are non-zero	Retain original for portfolio description; exclude status indic ators from recovery models
grade	All indicators match; one ac tive per row	Retain original; exclude encoded columns
sub_grade	All indicators match; one ac tive per row	Retain original; exclude encoded columns

home_ownership	All indicators match; one active per row	Retain original; exclude encoded columns
verification_status	All indicators match; one active per row	Retain original; exclude encoded columns
purpose	All indicators match; one active per row	Retain original; exclude encoded columns
addr_state	All indicators match; one active per row	Retain original for now; exclude encoded columns
initial_list_status	Both indicators match; one active per row	Retain original; exclude encoded columns

27. Unnamed 0 and Unnamed 0.1 are index. No useful data. The first column is a blank title, but is a duplicate. So will delete these 3 columns
28. Id and member id are not identical
29. Drop all the above mentioned columns that do not contribute. That would include the encodings, columns with 0, and first 3 unknown columns.

Data Cleaning Summary

Stage	Rows	Columns	Change
Original dataset (loan_data)	43,236	212	Imported from CSV
After Section 49 (loan_clean)	43,236	65	Removed 147 columns
After Section 54 (loan_clean)	43,236	57	Removed another 8 columns

1. Code Section 49 (147 cols dropped)

Category	Columns removed	Number	Reason
Index-like columns	First three columns of the original dataset	3	Redundant index/export fields
Completely empty	`annual_inc_joint`, `dti_joint`, `verification_status_joint`, `open_acc_6m`, `open_il_6m`, `open_il_12m`, `open_il_24m`, `mths_since_rcnt_il`, `total_bal_il`, `il_util`, `open_rv_12m`, `open_rv_24m`, `max_bal_bc`, `all_util`, `inq_fi`, `total_cu_tl`, `inq_last_12m`	17	Entirely missing; no usable information
Constant numeric/logical field	`good_bad`	1	All observations had value 0
One-hot encoded columns	All audited `grade:`, `sub_grade:`, `home_ownership:`, `verification_status:`, `loan_status:`, `purpose:`, `addr_state:`, and `initial_list_status:` indicator columns	126	Redundant with the corresponding original categorical variables

2. Section 54 (8 cols dropped)

Column	Reason for removal
`pymnt_plan`	Constant; all values were `"n"`
`out_prncp`	Constant; no variation
`out_prncp_inv`	Constant; no variation
`policy_code`	Constant; no variation
`application_type`	Constant; all values were `"INDIVIDUAL"`
`issue_d`	Redundant with `issue_d_date`
`earliest_cr_line`	Redundant with `earliest_cr_line_date`
`url`	Unique URL for each observation; not retained as a standard predictor

3. Date converted from string to date

Column(s)	Original format	Change made	Purpose
`issue_d_date`	Character (`YYYY-MM-DD`)	Converted to R `Date`	Enables date-based analysis
`earliest_cr_line_date`	Character (`YYYY-MM-DD`)	Converted to R `Date`	Enables date-based analysis

Dashboard Building

1. Load RDS and check summary
2. Built the frame for the dashboard with the main sections and the sub-sections.

Dashboard Structure

1. Part A - Portfolio
    a. Overview
    b. Segment Analysis
    c. Statistical Tests
    d. Relationships & Quality
2. Part B - Drivers
    a. Recovery Drivers
    b. Grade Validation
3. Part C - Models
    a. Recovery Probability
    b. Conditional Recovery
    c. LGD Estimates
    d. Model Performance
    e. LGD Simulator
4. Part D - Validation
    a. Robustness
    b. Portfolio Sensitivity
    c. Model Risk
5. Part E - Documentation
    a. Read First
    b. Business Report
    c. Detailed Report

PART A

Portfolio Overview and Recovery Characteristics

The prepared dataset contains 43,236 defaulted loan records. Positive recoveries were recorded for 24,371 loans (56.37%), whereas 18,865 loans (43.63%) had zero recorded recoveries. This indicates that the absence of recovery is a feature of the observed defaulted portfolio.

The mean loan-level recovery rate was approximately 6.08%, compared with a median of 2.95%. The higher mean is consistent with a right-skewed distribution in which relatively high recovery rates raise the average above the median. Consequently, the mean alone does not fully represent the recovery experience of a typical loan.

From a credit-risk management perspective, these findings support the probability of positive recovery from the amount recovered.

The recovery-rate measure is capped at 100% and does not incorporate the timing of recoveries, discounting or collection costs. The resulting LGD measure is therefore a baseline proxy rather than a fully discounted economic loss measure.

Segment Analysis of Recovery Performance

Loan grade displayed a notable pattern. The mean recovery rate increased from 4.63% for Grade A to 8.18% for Grade F, while Grade G recorded a mean recovery rate of 8.00%. This indicates that recovery severity among loans that have defaulted does not necessarily follow the same ordering as default risk.

Loan purpose also exhibited variation. Renewable energy loans recorded the highest mean recovery rate at 8.44%, although this segment contained only 52 observations. Debt consolidation represented the largest segment, with 26,127 loans and a mean recovery rate of 6.05%. The results suggest that recovery assumptions may differ across purposes, but estimates from small segments require particular caution.

Differences were also observed across loan terms. Loans with a 60-month term had a mean recovery rate of 6.98%, compared with 5.59% for 36-month loans. Their positive recovery incidence and mean recovery amounts were also higher. These results identify loan term as a potentially relevant recovery driver, although the differences may reflect other characteristics of the loans and borrowers.

Homeownership and employment length showed less consistent patterns. The mean recovery rates for borrowers classified as owners, mortgage holders and renters were relatively close, while the unusually high rate for the ‘NONE’ homeownership category was based on only eight observations. Employment-length categories also exhibited relatively modest differences, and the missing/unknown category recorded weaker recovery performance.

Statistical Testing of Recovery Performance

Objective

The statistical testing component was developed to assess whether recovery outcomes differ across borrower and loan segments. The analysis extends the descriptive comparisons performed in the Segment Analysis

section by providing formal statistical tests of differences in recovery performance and associations between segment characteristics and recovery incidence.

Methodology

The dashboard allows the user to select a grouping variable and an outcome variable. The available grouping variables are loan grade, loan purpose, homeownership, loan term and employment length. The outcomes include recovery rate, positive recovery status, recovery amount and an LGD proxy, defined as one minus the recovery rate.

A rule-based recommendation mechanism suggests a statistical test according to the selected outcome and the number of groups. The user can also select a test manually. The available tests are the Kruskal–Wallis test, Wilcoxon rank-sum test, Welch's analysis of variance (ANOVA), chi-square test of independence and Fisher's exact test.

For numeric outcomes, the Kruskal–Wallis test is recommended as an initial method when comparing three or more independent groups, given the highly skewed distribution of recovery rates. For two independent groups, the Wilcoxon rank-sum test is recommended. Welch's ANOVA is available when the objective is to compare group means without assuming equal population variances. For positive recovery status, the chi-square test of independence or Fisher's exact test can be used to investigate the association between the selected segment and whether a recovery was recorded.

The dashboard reports the test statistic, p-value, selected significance level and corresponding hypothesis decision. It also displays group-level sample sizes, means, medians and standard deviations, where applicable. These summaries help interpret the statistical results in the context of the observed portfolio.

Hypothesis Testing and Interpretation

For numeric outcomes, the null hypothesis is that there is no difference between the groups under the assumptions of the selected test. The alternative hypothesis is that at least one group differs. For positive recovery status, the null hypothesis states that recovery incidence is independent of the selected segment, while the alternative hypothesis states that an association exists.

The default significance level is 5%, although the user can select 1% or 10%. When the p-value is below the selected significance level, the dashboard indicates that the null hypothesis should be rejected. Otherwise, it indicates that the evidence is insufficient to reject the null hypothesis. Failure to reject the null hypothesis does not establish that the groups are identical.

Business Relevance

The statistical testing framework supports the evaluation of whether observed differences in recovery performance across loan segments are statistically distinguishable from random variation. This is relevant to loss-given-default (LGD) analysis because a single portfolio-wide recovery assumption may not adequately represent all segments.

The results can help identify segments for further investigation and inform the subsequent recovery-driver analysis and LGD modelling. Statistical significance alone, however, does not establish that a variable provides meaningful predictive value or that the difference is economically material. These considerations require effect-size assessment and multivariable modelling.

Limitations

The tests currently provide unadjusted comparisons and do not control for overlapping borrower and loan characteristics. Consequently, an observed association should not be interpreted as causal. The reliability of

the results also depends on the suitability of the selected test, independence of observations, group sizes and other relevant assumptions.

The automatic recommendation mechanism is rule-based and provides an initial test suggestion rather than a complete assessment of every statistical assumption. Although the implementation includes checks for incompatible outcome-test combinations and certain sparse-data conditions, further diagnostics and effect-size measures would strengthen the analysis. In addition, when a test indicates a significant difference across multiple groups, follow-up comparisons with an appropriate multiple-testing correction may be required to determine which groups differ.

Overall, the Statistical Tests component provides an interactive and reproducible framework for evaluating recovery differences across portfolio segments. Its results should be interpreted alongside descriptive statistics, business relevance and the subsequent multivariable modelling analysis.

Results for Statistical Tests

No.	Grouping variable	Outcome	Test	Statistic	df	p-value
1	Loan Grade	Recovery Rate	Kruskal–Wallis	671.2270	6	$< 2.2 \times 10^{-16}$
2	Loan Grade	Positive Recovery	Chi-square	154.5121	6	$< 2.2 \times 10^{-16}$
3	Loan Term	Recovery Rate	Wilcoxon rank-sum	189,394,137.5	—	$< 2.2 \times 10^{-16}$
4	Loan Purpose	Recovery Rate	Kruskal–Wallis	90.4512	13	1.146×10^{-13}
5	Employment Length	Positive Recovery	Chi-square	201.8288	11	$< 2.2 \times 10^{-16}$

Interpretation

1. Loan Grade and Recovery Rate

The Kruskal–Wallis test identified statistically significant differences in recovery-rate distributions across loan grades ((H = 671.227), (df = 6), (p < 2.2 \times 10^{-16})). The null hypothesis was rejected. Descriptive analysis showed that mean recovery rates ranged from 4.63% for Grade A to 8.18% for Grade F. This indicates that recovery severity varies across internal credit grades and supports further investigation of grade-based recovery segmentation.

2. Loan Grade and Positive Recovery

The chi-square test identified a statistically significant association between loan grade and positive-recovery status ((\chi^2 = 154.5121), (df = 6), (p < 2.2 \times 10^{-16})). The null hypothesis of independence was rejected. Therefore, the incidence of positive recovery is associated with internal credit grade. Combined with the recovery-rate result, this suggests that grade is relevant to both whether a recovery occurs and the distribution of recovery rates.

3. Loan Term and Recovery Rate

The Wilcoxon rank-sum test found a statistically significant difference in recovery-rate distributions between 36-month and 60-month loans ((W = 189{,}394{,}137.5), (p < 2.2 \times 10^{-16})). Descriptive statistics showed that the 60-month group had a mean recovery rate of 6.98%, compared with 5.59% for the 36-month group. The corresponding median recovery rates were 4.21% and 2.38%, respectively. These results indicate an association between loan term and recovery performance, although they do not establish that loan term causes the difference.

4. Loan Purpose and Recovery Rate

The Kruskal–Wallis test identified statistically significant differences in recovery-rate distributions across loan-purpose categories ((H = 90.4512), (df = 13), (p = 1.146 \times 10^{-13})). The null hypothesis was rejected. Descriptive results showed variation in mean recovery rates across purposes, including 8.44% for renewable-energy loans and 5.62% for credit-card loans. These results suggest that loan purpose may be useful for portfolio segmentation. However, the omnibus test does not establish that every pair of purpose categories differs significantly.

5. Employment Length and Positive Recovery

The chi-square test reported a statistically significant association between employment length and positive-recovery status ((\chi^2 = 201.8288), reported (df = 11), (p < 2.2 \times 10^{-16})). The reported degrees of freedom should be verified against the number of employment-length categories included in the test before this result is finalised. If the grouping variable and degrees of freedom are confirmed, the result would indicate that positive-recovery incidence varies across employment-length groups.

Overall Findings

All five tests reported statistically significant results at the 5% significance level. The findings indicate that recovery performance is not uniform across the examined loan and borrower segments. Internal credit grade is associated with both recovery incidence and recovery severity, while loan term and loan purpose are associated with differences in recovery-rate distributions. Employment length may also be associated with recovery incidence, subject to verification of the reported test configuration.

These findings support further segmentation and modelling of recovery outcomes. However, statistical significance alone does not establish practical importance or predictive value. The analysis consists of unadjusted comparisons, so observed differences may reflect overlapping characteristics such as loan amount, interest rate and borrower risk profile. The results should therefore be interpreted as evidence of association rather than causation. Pairwise post-hoc tests, effect-size measures and multivariable modelling would help assess the magnitude and robustness of the observed differences.

Correlation Analysis and Overlapping Information

A Pearson correlation heatmap was developed to examine linear relationships among selected numeric variables in the defaulted-loan portfolio. The interactive heatmap allows variables to be selected dynamically, while a complementary table identifies variable pairs with an absolute correlation of at least 0.70.

The observed correlation between funded amount and monthly instalment was particularly strong and positive ((r = 0.952)). This relationship is economically plausible because larger loans generally require

larger monthly instalments, although interest rates and loan terms also influence instalment amounts. Such strong correlation indicates overlapping information between the variables and warrants further investigation if both are considered as predictors in a regression model.

Correlation analysis is exploratory: it does not establish causality, and a high pairwise correlation alone does not prove multicollinearity. The results describe relationships within the available defaulted-loan dataset and should not automatically be generalised to all originated loans.

Data Quality Assessment

The prepared descriptive dataset contains 43,236 records and 57 variables. A total of 132,837 cells were missing, representing 5.39% of all dataset cells. This overall rate provides a useful summary but does not reveal whether missing values are concentrated in particular variables; variable-level missingness should therefore be considered when selecting modelling inputs and preprocessing methods.

No fully duplicated rows were detected. Recovery amounts contained no negative values, and all recovery rates fell within the defined interval of 0 to 1. These checks indicate that the principal recovery measures satisfy their specified numerical constraints. However, the absence of duplicate rows does not guarantee that loan identifiers are unique, and valid numerical ranges alone do not establish that every observation is economically plausible. Missing-data treatment and predictor selection should be documented separately, particularly where imputation is used for modelling.

PART B

Recovery Drivers

Objective

The objective of this analysis is to identify borrower and loan characteristics associated with the likelihood of obtaining recoveries after default. Understanding these relationships can help the institution assess recovery performance across different segments of its defaulted portfolio.

Methodology

The analysis uses the prepared descriptive dataset containing 43,236 defaulted loans. A loan is classified as having a positive recovery when its recorded recovery amount is greater than zero:

The percentage of loans with positive recoveries is calculated separately for each category of the selected characteristic. The dashboard also reports the mean and median recovery rates to provide additional information about recovery severity.

The dashboard allows comparisons across loan grade, loan purpose, homeownership, loan term, employment length, loan amount, interest rate, annual income, debt-to-income ratio, revolving credit utilisation, and recent delinquencies.

Findings and Interpretation

1. Loan grade

Recovery performance varies across internal grades. The descriptive analysis indicates that Grade F has a mean recovery rate of 8.18%, compared with 4.63% for Grade A. This suggests that recovery severity differs across grades, although these figures alone do not establish whether the grading system is effective.

2. Loan term

Loans with a 60-month term have a positive-recovery incidence of 58.5%, compared with 55.2% for loans with a 36-month term. The mean recovery rates are 6.98% and 5.59%, respectively. The differences suggest that loan term is associated with recovery outcomes in this defaulted portfolio.

3. Loan purpose

Recovery performance also varies by loan purpose. For example, renewable-energy loans have a mean recovery rate of 8.44%, while credit-card loans have a mean recovery rate of 5.62%. These comparisons should be interpreted cautiously because the number of loans differs substantially across purpose categories. The renewable-energy segment contains only 52 loans, whereas debt-consolidation loans account for 26,127 observations.

4. Homeownership and employment length

The descriptive results show differences in recovery performance across homeownership and employment-length categories. Such differences may reflect variation in borrower financial circumstances, asset ownership, repayment capacity, or other characteristics associated with the recovery process. However, these explanations are plausible mechanisms rather than causal conclusions established by the analysis.

Credit risk interpretation

Recovery outcomes may depend on borrower resources, the availability of assets that can be recovered, outstanding obligations, loan characteristics, and collection effectiveness. However, variables such as homeownership or income should not automatically be treated as direct measures of recoverable collateral. Their relationships with recovery outcomes may also reflect differences in the composition of borrowers and loans.

Conclusion

The analysis indicates that recovery outcomes are not uniform across borrower and loan segments. Loan grade, loan term, and loan purpose exhibit descriptive differences in recovery performance, while other selected characteristics can be investigated interactively using the dashboard.

These results are useful for identifying potential recovery drivers and prioritising further investigation. However, the analysis is descriptive and does not establish causality or isolate the independent contribution of each characteristic. Multivariable analysis would be required to assess whether these relationships persist after controlling for overlapping borrower and loan characteristics.

Furthermore, the dataset contains defaulted loans only. Consequently, the analysis concerns recovery performance conditional on default and cannot estimate the probability of default among all originated loans.

Grade Validation

Objective

The objective of Grade Validation is to evaluate whether the institution's internal grading system differentiates recovery performance among borrowers who have defaulted. Unlike the Recovery Drivers page, which examines several borrower and loan characteristics, this analysis focuses specifically on internal grades A through G.

Methodology

The analysis compares recovery outcomes across the seven internal grades using two complementary measures.

1. Positive-recovery incidence

For each grade, the positive-recovery rate is calculated as:

$$\text{Positive Recovery Rate}_g = \frac{\text{Number of loans with positive recoveries in grade } g}{\text{Total number of defaulted loans in grade } g} \times 100$$

This measure indicates how frequently loans in each grade generate any recovery after default.

2. Recovery severity

Mean and median recovery rates are calculated for each grade. The mean captures average recovery performance, while the median describes the middle observation and is less sensitive to unusually high recovery rates.

The dashboard also reports a mean LGD proxy, calculated as one minus the recovery rate. This represents a simplified loss measure and does not incorporate recovery timing, discounting, or collection costs.

Statistical Validation

Two statistical tests are used to assess whether the observed differences across grades are statistically significant.

1. **Chi-square test of independence**

This test evaluates whether internal grade and the occurrence of a positive recovery are statistically associated.

- Null hypothesis ((H_0)): Internal grade and positive-recovery incidence are independent.
- Alternative hypothesis ((H_1)): Internal grade and positive-recovery incidence are associated.

A statistically significant result indicates that positive-recovery incidence is not uniform across grades.

2. **Kruskal–Wallis test**

This non-parametric test evaluates whether recovery-rate distributions differ across the seven grades.

- Null hypothesis ((H_0)): Recovery-rate distributions are the same across grades.
- Alternative hypothesis ((H_1)): At least one grade has a different recovery-rate distribution.

The test is appropriate for investigating differences in recovery rates when the outcome is highly skewed. A significant result establishes that differences exist somewhere among the distributions, but does not identify every pair of grades that differs significantly.

Results

The descriptive analysis produces the following mean recovery rates:

Internal grade	Mean recovery rate
A	4.63%
B	5.19%
C	5.73%
D	6.35%
E	7.10%
F	8.18%
G	8.00%

The results indicate that mean recovery rates generally increase from Grade A through Grade F, before decreasing slightly for Grade G. Thus, the relationship is not perfectly monotonic: Grade G does not have the highest mean recovery rate.

The existing statistical tests from Part A provide further evidence of differences across grades:

- **Recovery-rate distributions:** Kruskal–Wallis statistic (H=671.227), with 6 degrees of freedom and ($p<2.2\times10^{-16}$).
- **Positive-recovery incidence:** Chi-square statistic (=154.5121), with 6 degrees of freedom and ($p<2.2\times10^{-16}$).

Both tests reject their respective null hypotheses at the 5% significance level. The Grade Validation dashboard independently calculates these tests from the dataset and displays the resulting statistics and p-values.

Interpretation

The findings provide evidence that internal grades differentiate recovery outcomes among defaulted borrowers. Both the frequency of obtaining any recovery and the distribution of recovery rates vary significantly across grades.

However, statistical significance does not automatically establish that the grading system is well calibrated, optimally ordered, or suitable for every recovery-related decision. The observed pattern in mean recovery rates is not perfectly monotonic, and the tests do not quantify the practical importance of the differences or identify all significant pairwise comparisons.

The analysis also does not establish whether the grades appropriately differentiate the probability of default. Since the dataset contains only defaulted loans, it evaluates recovery performance conditional on default rather than default-risk differentiation across the full originated portfolio.

Conclusion

The Grade Validation analysis finds statistically significant differences in both positive-recovery incidence and recovery-rate distributions across internal grades. This indicates that the grades contain information associated with recovery outcomes among defaulted borrowers.

Note

A higher recovery rate should not be interpreted as indicating a better internal credit grade. Assuming the conventional ordering in which Grade A represents lower credit risk and Grade G represents higher credit risk, the observed differences describe recovery performance conditional on default rather than overall credit quality.

Grade F has the highest mean recovery rate (8.18%), followed by Grade G (8.00%), while Grade A has the lowest (4.63%). These results do not establish that the grade ordering is optimal or that higher-risk grades are preferable. Instead, they indicate that recovery outcomes vary across grades, warranting further investigation into the factors underlying these differences.

PART C

Level of Missingness

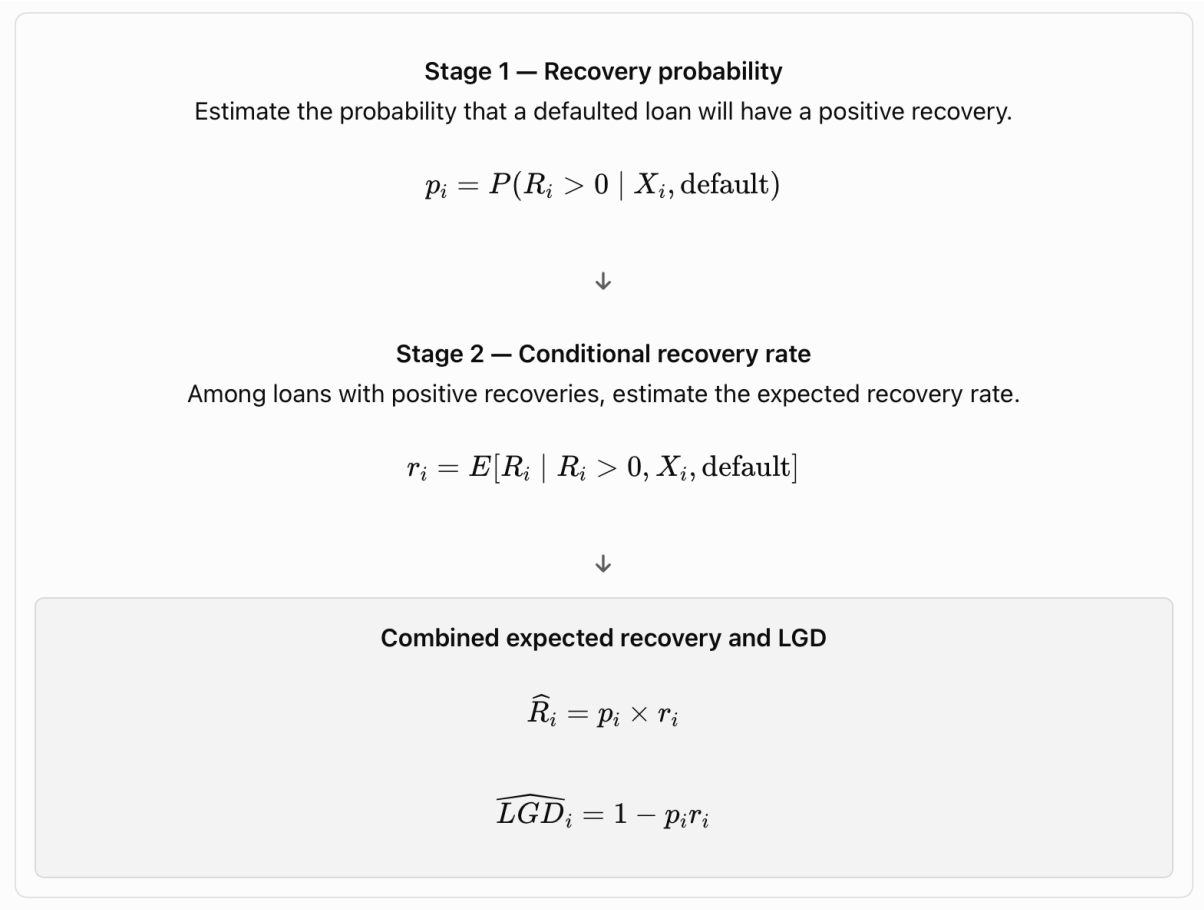
Variable(s)	Missing	Recommended treatment for Part C
next_pymnt_d	98.24 %	Exclude. It is also a post-default/payment-related field.
mths_since_last_major_derog	81.61 %	Exclude from the initial model; its extensive missingness makes it difficult to use reliably.
desc	63.43 %	Exclude free-text descriptions from the initial model.
tot_coll_amt, tot_cur_bal	24.93 % each	Retain as candidate predictors if they are available at the chosen prediction date; use the existing imputation and missingness indicators.
emp_title	7.60%	Exclude from the initial model because it is high-cardinality free text.
emp_length	5.41%	Use the prepared emp_length_model variable, which represents missing employment length as Unknown.
last_pymnt_d	0.87%	Exclude because it is payment-history information recorded after origination.
revol_util	0.12%	Retain as a candidate predictor; use the existing median imputation.
collections_12_mths_ex_med	0.06%	It was removed from the modelling dataset already; that is reasonable given the earlier preparation decision.
last_credit_pull_d	0.01%	Exclude from the initial model because it is a later credit-report date.
title	0.01%	Exclude from the initial model; it is largely a descriptive text field.
earliest_cr_line_date	0.01%	Prefer the derived mths_since_earliest_cr_line variable rather than the raw date.

Variables that can cause data leakage

Variable	Reason for exclusion
recoveries	Direct recovery amount; used to construct the outcome.
recovery_rate	Recovery target used in Stage 2 and in the combined LGD calculation.
CCF	Potentially derived from exposure and recovery-related information; exclude until its exact definition and timing are verified.
total_pymnt	Cumulative payments may include payments received after default.

total_pymnt_inv	Investor-level cumulative payments may include post-default outcomes.
total_rec_prncp	Principal repaid is a realised payment outcome.
total_rec_int	Interest received is a realised payment outcome.
total_rec_late_fee	Realised late-fee collections.
collection_recovery_fee	Directly related to recovery and collection activity.
last_pymnt_d	Records the date of the last payment; may reflect post-default activity.
last_pymnt_amnt	Records a realised payment amount.
next_pymnt_d	Payment-related date that may be recorded after the prediction point.
last_credit_pull_d	Credit-report date that may postdate default.

2 Stage Modelling Approach



The following two targets:

- Stage 1 — Recovery Probability: positive_recovery = (recoveries > 0), using all 43,236 loans.
- Stage 2 — Conditional Recovery: recovery_rate, using only the 24,371 loan with positive recoveries

The combined LGD estimate will be:

$$\widehat{LGD}_i = 1 - \left(\widehat{P}(R_i > 0 \mid X_i) \times \widehat{E}[R_i \mid R_i > 0, X_i]\right)$$

The 19 variables being used for the model:

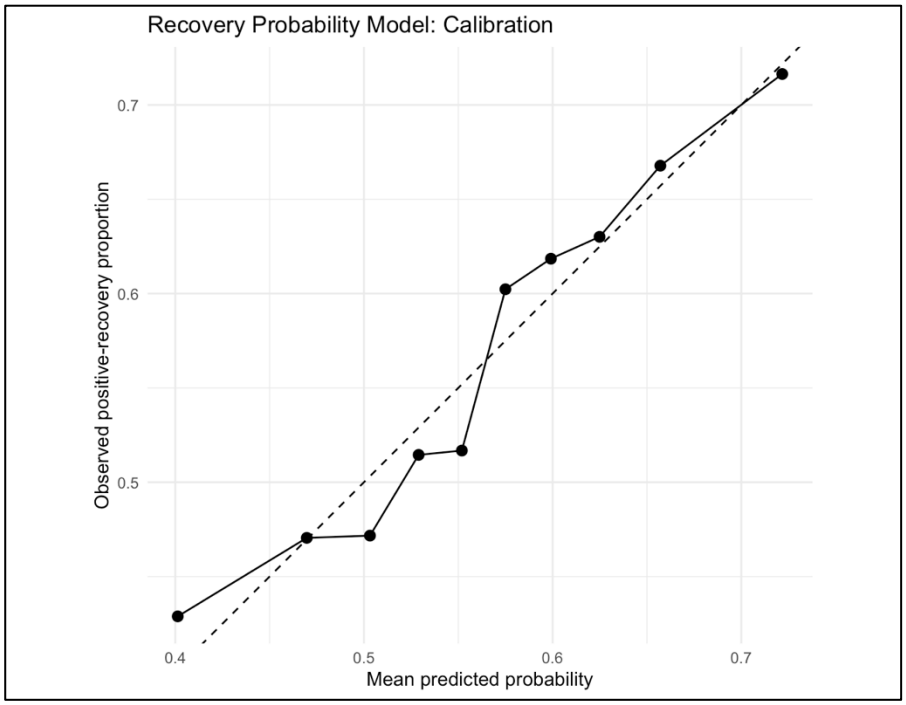
Variable	Initial decision	Reason
funded_amnt	Include	Exposure denominator for the recovery-rate definition.
term	Include	Loan term.
int_rate	Include	Contractual interest rate.
grade	Include	Internal risk grade, if assigned by the prediction date.
purpose	Include	Loan purpose.
home_ownership	Include	Borrower characteristic.
annual_inc	Include	Borrower income.
verification_status	Include	Income verification status.
emp_length_model	Include	Employment length with missing values represented as Unknown.
dti	Include	Debt-to-income ratio.
delinq_2yrs	Include	Recent delinquency history.
inq_last_6mths	Include	Recent credit inquiries.
open_acc	Include	Open credit accounts.
pub_rec	Include	Public-record indicator.
revol_bal	Include	Revolving balance.
revol_util	Include	Revolving utilisation; existing imputation should be retained.
total_acc	Include	Total credit accounts.
mths_since_earliest_cr_line	Include	Length of credit history, expressed as a derived numeric measure.
acc_now_delinq	Include	Delinquency indicator, provided it reflects status at default.

Stage 1 : Recovery Probability

Model 1 of 2: Logistic Regression Model

Metric	Result	Interpretation
ROC-AUC	0.6061	Weak discrimination; only modestly above 0.5
Brier score	0.2378	Probability prediction error; needs comparison with a baseline
Log loss	0.6682	Probability quality; lower is better
Accuracy	58.50%	Only a modest improvement over always predicting the majority class
Precision	59.78%	Of loans predicted to recover, about 60% actually had positive recoveries
Recall	80.66%	The model identifies about 81% of loans with positive recoveries
Specificity	29.87%	It identifies fewer than one-third of zero-recovery loans
F1 score	0.6866	Reflects the trade-off between precision and recall

The baseline logistic regression is working, but its results show that it has limited ability to distinguish loans with positive recoveries from those with zero recoveries. We should retain it as a benchmark and test wheth er Random Forest improves performance.



The calibration results are more encouraging than the discrimination results.

- In the lowest probability group, the mean prediction was 40.13%, while the observed positive-recovery rate was 42.89%.
- In the highest probability group, the prediction was 72.17%, versus an observed rate of 71.64%.
- The middle groups show some deviations, but observed recovery frequency generally increases with predicted probability.

This suggests that the model captures the broad relationship between predicted probability and observed rec overy frequency, although it does not rank individual loans particularly well. Calibration across ten groups is only a diagnostic; it does not establish that probabilities are fully calibrated.

Model 2 of 2: Random Forest Model (500 trees)

Result was only slightly better than logistic.
Tuning using 3 fold cross validation on training set.
Best : mtry = 3 , node size = 5

Final tuned Random Forest results

Metric	Test result
ROC-AUC	0.6252
Brier score	0.2344
Log loss	0.6608
Accuracy	60.25%
Precision	61.24%
Recall	80.31%
Specificity	34.32%
F1 score	0.6949

Interpretation

- Discrimination remains weak: ROC-AUC of 0.6252 indicates limited ability to distinguish positive-recovery loans from zero-recovery loans.
- Recall is relatively high: The model correctly identifies 80.31% of loans that have positive recoveries
- Specificity is low: It correctly identifies only 34.32% of loans with zero recoveries.
- Classification is imbalanced: At the 0.5 threshold, the model predicts positive recovery for many loa ns that ultimately have zero recovery.
- Probability metrics: The Brier score and log loss will be useful for comparing models, but we should also examine calibration before deciding which probabilities to use in the LGD framework.

Confusion matrix interpretation

Actual outcome	Predicted zero recovery	Predicted positive recovery
Zero recovery	1,295	2,478
Positive recovery	960	3,915

Of the 4,875 loans with positive recoveries in the test set, the model correctly identifies 3,915. However, it i ncorrectly classifies 2,478 of the 3,773 zero-recovery loans as having positive recovery.

Comparison against logistic regression

Metric	Logistic regression	Tuned Random Forest
ROC-AUC	0.6061	0.6252
Brier score ↓	0.2378	0.2344
Log loss ↓	0.6682	0.6608
Accuracy	58.50%	60.25%
Recall	80.66%	80.31%
Specificity	29.87%	34.32%
F1 score	0.6866	0.6949

The tuned Random Forest performs better on most metrics, while logistic regression has marginally higher recall. The improvements are modest, so we should not describe the Random Forest as a highly accurate model.

Checking Results in Deciles group

Decile	Predicted	Observed	Interpretation
1	38.77%	39.42%	Slight underprediction
2	45.75%	41.39%	Overprediction
3	49.49%	50.06%	Close
4	52.40%	52.83%	Close
5	55.15%	52.60%	Overprediction
6	57.82%	57.11%	Close
7	60.44%	63.24%	Underprediction
8	63.37%	64.97%	Underprediction
9	66.97%	67.94%	Close
10	74.04%	74.19%	Very close

Main findings

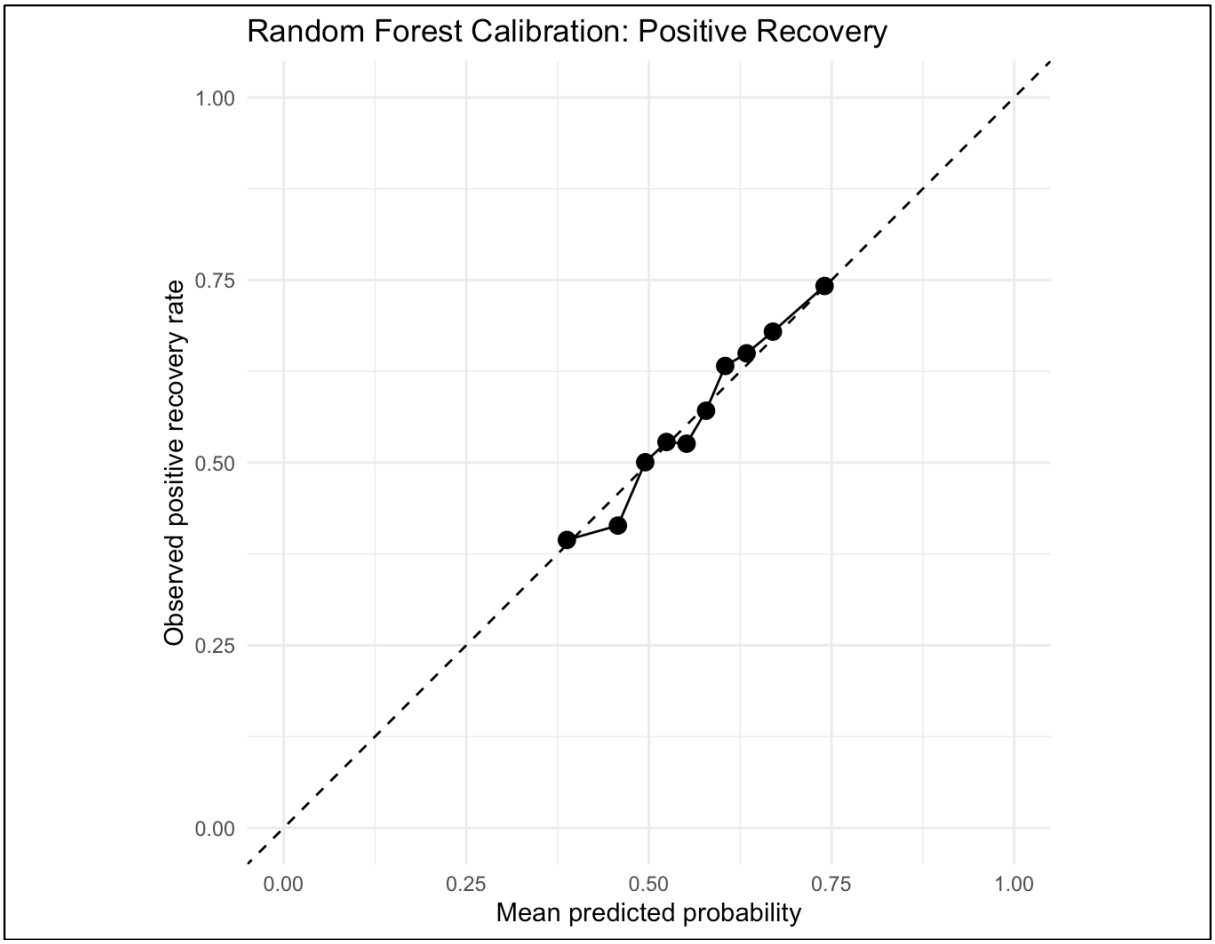
- The highestprobability decile is well calibrated: predicted recovery probability is 74.04%, compared with an observed rate of 74.19%.
- The second decile shows the largest deviation, with predicted probability exceeding the observed rate by 4.36 percentage points.
- The model underpredicts recovery in deciles 7–9, most notably in decile 7.
- Overall, calibration is encouraging, but the decile results do not establish perfect calibration or guara ntee accuracy for individual loans.

Stage 1 conclusion

We now have a preliminary selection for the recovery-probability model:

- Selected candidate: tuned Random Forest (mtry = 3, nodesize = 5, 500 trees).
- Test ROC-AUC: 0.6252.
- Test Brier score: 0.2344.
- Test log loss: 0.6608.
- Calibration: broadly reasonable across deciles, with some local deviations.

Model's probabilities appear reasonably aligned with observed rates in groups, but its ability to discriminate between positive- and zero-recovery loans remains limited. We should describe both findings in the report.



Stage 2: Conditional Recovery Rate

Model 1 of 2: Linear Regression

Compare the baseline

Diagnostic	Result	Interpretation
Test MAE	0.0521	Average absolute error of about 5.21 percentage points
Test RMSE	0.0947	Larger errors have a greater influence on this metric
Training \(R^2\)	0.0429	The model explains about 4.29% of the training variation
Predictions outside [0, 1]	0	No predictions needed clipping in this test run

The linear model's explanatory power is low. Its coefficient significance does not necessarily translate into st rong predictive performance. The maximum residual of approximately 0.93 also indicates that some positive -recovery loans have very large prediction errors.

Model 2 of 2: Random Forest

Training using 3 fold cross validation (500 trees)
Mtry – 3,4,6 and node size – 1,5,10

The Stage 2 tuned Random Forest performs slightly better than the linear regression baseline on both test me trics. We can now use it as the selected candidate for estimating conditional recovery rates.

Final Stage 2 comparison

Metric	Linear regression	Tuned Random Forest
Test MAE	0.0521	0.0515
Test RMSE	0.0947	0.0938
mtry	—	3
nodesize	—	10
Trees	—	500

The Random Forest reduces MAE by approximately 1.15% and RMSE by approximately 0.95% relative to l inear regression. These are modest improvements, not a large performance difference.

Combining Stage 1 and Stage 2 to make the LGD Estimator

Results

Metric	Estimated LGD	Observed LGD proxy
Mean	93.77%	93.94%
Median	93.89%	97.28%

Prediction metric	Result
Test observations	8,648
LGD MAE	0.0622
LGD RMSE	0.0889

Result Interpretation

- Mean LGD: The model's mean estimate is 0.17 percentage points below the observed mean. This suggests low aggregate bias on this test sample.
- Median LGD: The predicted median is lower than the observed median by 3.39 percentage points. The model may not capture the distribution of individual loan-level outcomes well, despite the close means.
- MAE: The average absolute error is approximately 6.22 percentage points.
- RMSE: The root mean squared error is approximately 8.89 percentage points. Because RMSE penalizes larger errors more heavily, its value being higher than MAE suggests that larger individual errors affect performance.

Running Diagnostics

These diagnostics reveal an important distinction i.e. the two-stage Random Forest estimates the portfolio's average LGD reasonably closely, but its ability to distinguish between individual loans is weak.

Diagnostic	Result	Interpretation
Mean prediction error (bias)	−0.18 percentage points	Very small average overprediction of recovery loss? No—the negative sign means estimated LGD is slightly below observed LGD on average.
Median prediction error	−3.41 percentage points	The typical signed error is negative.
Median absolute error	5.60 percentage points	Half of loans have absolute errors no greater than 5.60 percentage points.
90th percentile absolute error	8.84 percentage points	90% of loans have absolute errors no greater than 8.84 percentage points.
95th percentile absolute error	11.57 percentage points	The largest 5% of errors exceed approximately 11.57 percentage points.
Predicted–observed correlation	0.1272	Weak linear association between predicted and observed loan-level LGD.

Comparing our model to the mean benchmark

Metric	Two-stage Random Forest	Mean benchmark	Improvement
MAE	0.0622	0.0633	1.74% lower
RMSE	0.0889	0.0895	0.67% lower

The relative improvements are calculated against the benchmark's MAE and RMSE.

Interpretation

- The Random Forest improves on both metrics, so it is not merely reproducing a constant portfolio-average LGD.
- The improvement is small. The model captures some variation in LGD, but the correlation of 0.1272 indicates that loan-level predictions remain weakly associated with realised LGD.
- The model's close aggregate mean should not be confused with strong individual prediction accuracy.

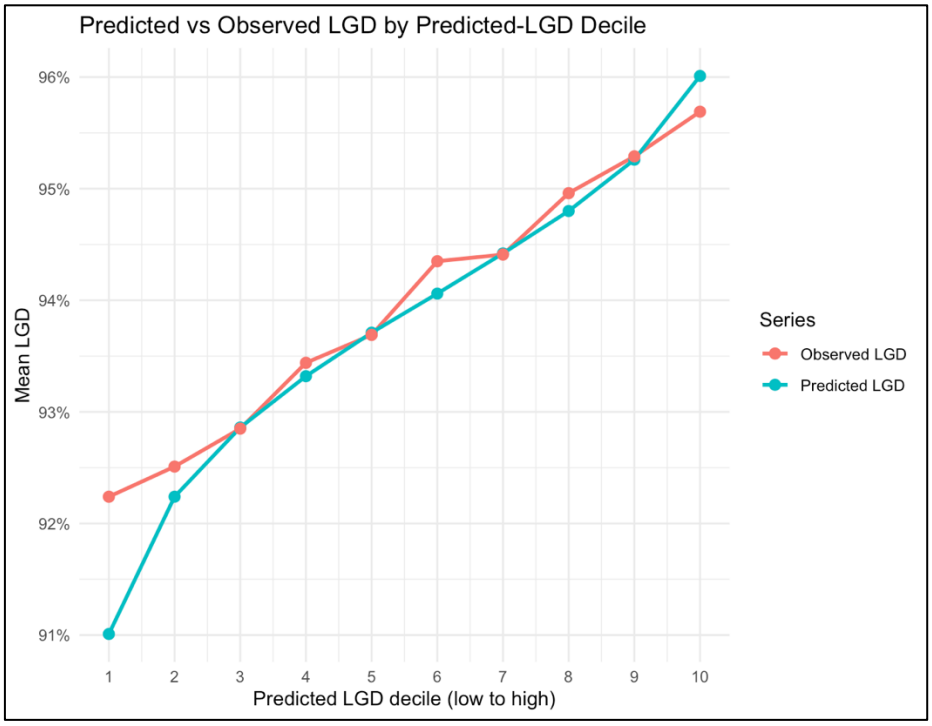
Checking model performance at the decile level

Finding	Evidence
Risk ordering is preserved	Mean observed LGD increases from 92.24% in decile 1 to 95.69% in decile 10.
Predictions are close within groups	The absolute difference between mean predicted and observed LGD is below 1.23 percentage points in every decile.
Some local miscalibration remains	Decile 1 underpredicts LGD by 1.23 percentage points; decile 10 overpredicts it by 0.32 percentage points.
The model has limited separation	The observed LGD range between the lowest and highest deciles is only 3.45 percentage points.

The model is reasonably calibrated at the decile level, but its ability to differentiate individual loans is weak. These findings are not contradictory. Group averages can be close even when individual predictions are noisy.

Summary for Part C

- Stage 1 estimates the probability of positive recovery.
- Stage 2 estimates the conditional recovery rate.
- Their product gives expected recovery rate, and one minus that value gives estimated LGD.
- The combined model marginally outperforms the training-mean benchmark on MAE and RMSE.
- Mean LGD is close to the observed test-set mean.
- Decile calibration is close, but the predicted-versus-observed correlation is low.



The decile results confirm that the two-stage model has good group-level calibration but limited differentiation between loans. The predicted and observed LGD values track each other closely across all ten deciles.

Interpretation of the decile plot

- Decile 1: LGD is underpredicted by 1.23 percentage points.
- Deciles 2–9: Differences are very small, with absolute discrepancies no greater than 0.29 percentage points.
- Decile 10: LGD is overpredicted by 0.32 percentage points.
- Overall: The model preserves the ordering of observed LGD across deciles, from 92.24% in the lowest predicted-LGD decile to 95.69% in the highest.

Distribution comparison between LGD observed and predicted

Metric	Predicted LGD	Observed LGD proxy
Mean	93.77%	93.94%
Median	93.89%	97.28%
Standard deviation	1.45 pp	8.96 pp
Minimum	78.88%	0.00%
Maximum	97.87%	100.00%

Where pp = percentage points.

The key findings

- Aggregate estimation is close: the difference between mean predicted and observed LGD is just 0.17 percentage points.
- The median is underestimated: predicted median LGD is 3.39 percentage points below the observed median.
- Predictions are compressed: predicted LGD's standard deviation is 1.45 percentage points, compared with 8.96 percentage points for observed LGD. The model's predicted spread is only about 16% of the observed spread.
- Extreme outcomes are poorly represented: the observed LGD proxy ranges from 0% to 100%, while predictions range from 78.88% to 97.87%.

This compression is consistent with the weak loan-level correlation of 0.1272 and the small improvement over the constant-mean benchmark. The model tends to predict LGD values near the portfolio average instead of reproducing the full range of realised outcomes.

Conclusion

These diagnostics indicate that the average error is small, but individual loan-level errors can be materially larger. In particular, the low correlation and compressed predicted distribution suggest that the model tends to produce similar LGD estimates for loans whose observed outcomes differ substantially.

This behaviour is consistent with a model that captures broad recovery patterns but does not fully capture the heterogeneity of realised recovery outcomes. Random Forest models can average predictions across

terminal nodes, which may contribute to less variable predictions, particularly when the predictors provide limited information about individual outcomes.

Accordingly, the combined framework should not be evaluated solely on the basis of average error. Its ability to distinguish between loans with different realised LGD values is also important for practical risk-management applications.

Business Implications

First, loan-level LGD estimates can support the assessment of expected loss severity after default. When combined with an appropriate exposure-at-default measure, they can contribute to loss calculations and recovery planning.

Second, the separation of recovery occurrence from conditional recovery severity helps distinguish loans that are unlikely to generate any recovery from loans for which positive recovery is expected.

Third, the estimates can support portfolio-level comparisons and sensitivity analysis. However, the weak relationship between predicted and observed LGD means that the current model should be used cautiously for decisions requiring precise loan-level loss differentiation.

The results also demonstrate why a model that performs reasonably well on average may still have limitations when applied to individual accounts. Further improvements would be necessary before relying on these estimates for high-impact operational or regulatory decisions.

PART D

1 of 2: Robustness

Finding	Interpretation
LGD MAE = 0.0622	Average absolute error is approximately 6.22 percentage points.
Benchmark MAE = 0.0633	The model improves MAE by only about 1.74% relative to the benchm ark.
Prediction correlation = 0.1272	Predicted LGD has a weak linear relationship with observed LGD.
Predicted LGD SD = 0.0145; ob served SD = 0.0896	The model's predictions are much less dispersed than actual outcomes.
Decile calibration	Mean predicted and observed LGD are relatively close within most dec iles, but this does not establish strong loan-level predictive accuracy.

Robustness by loan grade

Grade	Loans	MAE	RMSE	Mean error
A	497	0.0531	0.0945	+0.0024
B	1,814	0.0561	0.0848	−0.0018
C	2,419	0.0615	0.0914	−0.0004
D	2,032	0.0637	0.0832	−0.0036
E	1,171	0.0670	0.0826	−0.0064
F	566	0.0752	0.1161	+0.0077
G	149	0.0713	0.0796	−0.0108

Mean error = predicted LGD − observed LGD. Positive values indicate average overestimation; negative val ues indicate average underestimation. Values are in LGD proportion units.

The results show that prediction errors differ across credit grades. Grade F has the highest MAE and RMSE, indicating comparatively weaker predictive accuracy for this segment. Grade G has a relatively high MAE and a smaller sample of 149 loans, so its results should be interpreted cautiously.

The mean prediction errors also indicate differences in the direction of error. The model tends to overpredict LGD for grades A and F and underpredict it for grades B, C, D, E, and G in this test set. These are descriptive findings; no statistical significance tests were conducted to establish whether the differences persist beyond this sample.

Predicted LGD variability is also much lower than observed variability within each grade. This reinforces the finding that the framework does not fully capture loan-level outcome dispersion, even when the portfolio is segmented by grade.

Consequently, the overall test-set performance should not be assumed to apply uniformly to all credit grades. Segment-level monitoring is necessary to identify groups for which model performance deteriorates or systematic prediction errors emerge.

Uncertainty in Performance Estimates

Bootstrap resampling of the existing test-set prediction and observation pairs was used to estimate uncertainty around the combined framework's error metrics. The analysis used 1,000 bootstrap resamples.

Bootstrap results

Metric	Estimate	95% bootstrap confidence interval
MAE	0.06220	[0.06096, 0.06354]
RMSE	0.08889	[0.08383, 0.09425]

Interpretation

Under resampling of the existing test observations, the estimated MAE lies between approximately 6.10 and 6.35 percentage points, while RMSE lies between approximately 8.38 and 9.43 percentage points. These inte rvals quantify uncertainty from test-
sample composition, conditional on the fitted model; they do not establish performance on future portfolios.

The bootstrap intervals indicate the range of error estimates obtained through resampling the existing test-set observations. They provide information about sampling variability conditional on the fitted models and their existing predictions.

These intervals do not incorporate the additional uncertainty that could arise from retraining the models, selecting different hyperparameters, changes in the training sample, or shifts in future portfolio characteristics. They should therefore not be interpreted as complete measures of model uncertainty or as guarantees of future performance.

The analysis provides evidence about the stability of the error estimates under resampling, but additional validation would be required to establish robustness under changing economic or portfolio conditions.

Part 2 of 2: Portfolio sensitivity

If the institution holds a larger proportion of lower-grade loans and a smaller proportion of higher-grade loans, how much does its portfolio-level estimated LGD change?

Objective

The objective of this analysis is to assess how changes in portfolio composition affect aggregate estimated Loss Given Default (LGD) and the implied recovery rate. Since the developed framework estimates LGD conditional on default, the analysis focuses on recovery outcomes among defaulted loans rather than the probability of default.

Methodology

The existing test-set predictions were matched to the corresponding loan records using their original row indices. Loans were grouped into seven credit grades, A through G, and the mean predicted LGD and mean observed LGD were calculated for each grade.

Three portfolio-composition scenarios were evaluated:

1. **Baseline:** The original grade distribution in the test dataset.
2. **More A–C:** A 10-percentage-point increase in the combined portfolio share of grades A–C, funded by a proportional reduction in the combined share of grades E–G.
3. **More E–G:** A 10-percentage-point increase in the combined share of grades E–G, funded by a proportional reduction in the combined share of grades A–C.

The share of grade D was held constant. Within each group, the original relative proportions of the constituent grades were preserved. For each scenario, aggregate estimated LGD was calculated as the weighted average of the grade-level mean predicted LGDs. The corresponding implied recovery rate was calculated as one minus estimated LGD.

This is a controlled composition scenario: grade-level predictions remain fixed, and only the proportions of loans assigned to each grade group change.

Results

Scenario	Estimated LGD	Implied recovery rate	Change in LGD vs. baseline
Baseline	93.7686%	6.2314%	—
More A–C	93.9771%	6.0228%	+0.2086 percentage points
More E–G	93.5600%	6.4400%	−0.2086 percentage points

Under the baseline composition, the model estimates portfolio LGD at approximately 93.77%, corresponding to an implied recovery rate of 6.23%. Increasing the combined share of grades A–C by 10 percentage points raises estimated LGD to approximately 93.98%. Conversely, increasing the combined share of grades E–G by the same amount reduces estimated LGD to approximately 93.56%.

The observed pattern reflects the grade-level recovery relationships in this particular dataset. The mean recovery rate in the prepared dataset increases from approximately 4.63% for grade A to 8.18% for grade F, before decreasing slightly to 8.00% for grade G. In the test subset, mean observed LGD generally decreases from grade A to grade F, with a higher value for grade G than for grade F. Thus, the model's estimated portfolio LGD is sensitive to the grade mix, but the direction of this sensitivity should not be interpreted as a universal ranking of credit risk.

The baseline weighted estimate reproduces the existing aggregate predicted LGD of approximately 93.77%. The weighted observed-LGD reference is approximately 93.94%, giving a prediction difference of about −0.18 percentage points. This provides a consistency check against the existing test-set results.

Business Implications

The analysis demonstrates that portfolio composition can influence aggregate LGD estimates even when the underlying model and grade-level predictions remain unchanged. Portfolio monitoring should therefore consider changes in the distribution of defaulted loans across grades, rather than relying exclusively on a single portfolio-wide LGD estimate.

The scenario results can support portfolio sensitivity reporting and help illustrate the potential effect of changes in grade composition on estimated recoveries. However, the results should be interpreted in the context of the dataset's observed recovery patterns. The grade labels alone do not establish that one group is economically safer than another for recovery purposes.

Limitations

The analysis uses the existing test-set predictions and observed outcomes, rather than a newly sampled or independently validated portfolio. The scenarios are illustrative and do not represent forecasts of actual future portfolio composition.

Within-grade loan characteristics and predicted outcomes are held fixed. Consequently, the analysis does not capture changes in borrower characteristics within grades, interactions between portfolio composition and model performance, or changes in recovery conditions over time. The analysis is conditional on default and does not incorporate probability of default or exposure at default into an expected-credit-loss estimate.

Overall, the portfolio sensitivity analysis shows that the estimated aggregate LGD changes when the grade composition changes. These results support the use of composition monitoring and scenario analysis as supplementary tools for understanding LGD model outputs, while recognising the limitations of the underlying dataset and modelling assumptions.

CONCLUSION
Assumptions, Limitations, and Sources of Model Risk

The following considerations affect the interpretation and potential use of the framework.

1. Dataset scope and selection effects

The analysis is based on defaulted loans in the available dataset. The resulting LGD estimates are conditional on default and should not be generalised automatically to all originated loans or other lending

portfolios. Differences in loan characteristics, recovery practices, borrower populations, and economic conditions may limit the transferability of the results.

2. Two-stage modelling assumptions

The framework combines the estimated probability of positive recovery with the conditional recovery-rate estimate. Its reliability depends on both components. Errors in either stage can affect the final LGD estimate, and the conditional recovery model does not independently represent loans with zero recovery.

3. Prediction compression

The low variability of predicted LGD relative to observed LGD, together with the weak predicted-versus-observed correlation, indicates that the framework does not capture the full dispersion of individual outcomes. This creates a risk that estimates for particular loans may be materially different from realised outcomes even when the portfolio average is reasonably close.

4. Segment-level model risk

The differences in errors across grades indicate that performance is not uniform. A model that appears acceptable at portfolio level may be less reliable for particular segments. The smaller number of grade G observations also limits confidence in the stability of its segment-level metrics.

5. Temporal and population stability

The reported validation uses an existing held-out test set and bootstrap resampling of its predictions. No out-of-time validation result is available in this assessment. Consequently, the analysis does not establish how the framework would perform under changing economic conditions, recovery policies, or portfolio composition.

6. Predictor availability and target leakage

The selected predictor list excludes direct recovery outcomes such as recoveries and recovery rate. However, this alone does not establish that every predictor is appropriate at the intended prediction date. Predictor availability should be verified against the operational use case, and the modelling process should exclude information that would only become available after the prediction date.

7. Dependence on historical patterns

The Random Forest models learn relationships present in the historical data. Changes in recovery processes or borrower characteristics may weaken those relationships. The existing results do not quantify the effect of such changes, so ongoing monitoring and independent validation are necessary.